

3(a)	<u>sum</u> / <u>total</u> momentum before = <u>sum</u> / <u>total</u> momentum after or <u>sum</u> / <u>total</u> momentum (of a system of objects) is constant	M1
	if no (resultant) external force / for an isolated system	A1
3(b)(i)	$3m \times 4 = m \times v \sin \theta$ ($v \sin \theta = 12$)	C1
	$2m \times 6 = m \times v \cos \theta$ ($v \cos \theta = 12$)	C1
	therefore $\sin \theta = \cos \theta$ or $\tan \theta = 1$ $\theta = 45^\circ$	A1
3(b)(ii)	$mv \times \cos 45^\circ = 12m$ or $mv \times \sin 45^\circ = 12m$ or $(mv)^2 = (3m \times 4)^2 + (2m \times 6)^2$	C1
	$v = 17 \text{ m s}^{-1}$	A1
3(c)(i)	(chemical energy) = $0.0050 \times 700 = 3.5 \text{ (J)}$ or (chemical energy) = $5.0 \times 0.700 = 3.5 \text{ (J)}$	A1

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Question	Answer	Marks
3(c)(ii)	$E = \frac{1}{2}mv^2$	C1
	total $E = (0.5 \times 3m \times 4^2) + (0.5 \times 2m \times 6^2) + (0.5 \times m \times 17^2)$	C1
	$3.5 = 204m$ $m = 0.017 \text{ kg}$	A1

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